

VENKATA DUGGIRALA

SENIOR / LEAD QA AUTOMATION ENGINEER | SDET | QUALITY ENGINEERING | AUTOMATION ARCHITECTURE | CI/CD | CLOUD & KUBERNETES

San Jose, CA | 408-250-9952 | venkata.duggirala@gmail.com | linkedin.com/in/venkataduggirala

PROFESSIONAL SUMMARY

Senior Quality Engineering and Automation leader with 20+ years of engineering experience spanning software development, test automation architecture, CI/CD, cloud infrastructure, cybersecurity, distributed systems, payments, telecommunications, and semiconductor/EDA technologies. Proven record of architecting scalable automation frameworks, embedding shift-left quality controls into CI/CD, automating Kubernetes-based environments, validating APIs and microservices, and driving performance, scale, security, and release engineering improvements. Combines hands-on engineering depth with cross-functional leadership across Python, pytest, Selenium, REST APIs, Pact, Jenkins, CloudBees, Harness, Kubernetes, Azure, OpenStack, SonarQube, Snyk, JMeter, Node.js, MongoDB, Mocha, C++, Qt, and Perl.

CORE TECHNICAL SKILLS

Automation & Quality Engineering: Test Automation Architecture, pytest, BDD, UI Automation, API Automation, Pact Contract Testing, Microservices Testing, Integration Testing, End-to-End Testing, Regression Engineering, Performance & Load Testing, Scale Testing, Mobile Testing, Accessibility Testing, Test Strategy, Release Validation

CI/CD, DevOps & Cloud: Harness, Jenkins, CloudBees, Kubernetes, Azure, OpenStack, GitHub, Bitbucket, SonarQube Quality Gates, Snyk Vulnerability Scanning, Containerized Test Automation, Dynamic Test Environments, OVF, qcow2, vApp

Programming & Frameworks: Python, Selenium WebDriver, REST Automation, Node.js, Mocha, JMeter, MongoDB, C++, Qt, Perl, Frank/Cucumber

Leadership & AI-Assisted Engineering: Automation Strategy, Architecture, Shift-Left Quality, DevSecOps, Technical Mentoring, Hiring, Cross-Functional Leadership, Customer Escalations, POC Support, Claude AI, Windsurf AI

PROFESSIONAL EXPERIENCE

PROOFPOINT | Senior QA Automation Engineer

Nov 2022 – Present

Sunnyvale, CA

- Spearheaded CI/CD modernization, accelerating release cycles from monthly to weekly (and twice-weekly for key services) by embedding SonarQube, Snyk scans, and test automation into CloudBees and Harness pipelines.
- Elevated engineering quality across Dev and QA, mentoring teams on robust test design and directly resolving critical blockers across dev cycles, staging, and high-severity production incidents.
- Authored comprehensive engineering wikis and technical documentation, establishing organization-wide best practices, automation standards, and execution blueprints for QA and process automation.
- Architected containerized microservice test frameworks on Kubernetes using Python, pytest, Playwright, and Pact contract testing, catching breaking API changes early.
- Orchestrated multi-region test automation for global rollouts across India, Australia, Canada, and UAE, running daily automated runs with zero-downtime reliability.
- Expanded test authoring org-wide by building a reusable pytest BDD layer, enabling Product and Support teams to write scenarios and boosting coverage without added QA headcount.
- Engineered a unified test reporting dashboard to aggregate and publish real-time results from heterogeneous frameworks and languages into a single view.
- Eliminated production bottlenecks by running REST API load tests that surfaced critical memory spikes and scalability issues pre-release.
- Drove key infrastructure migrations, shifting repos from Bitbucket to GitHub, pipelines from Jenkins to CloudBees, and fixing broken unit tests and dependency vulnerabilities.
- Leveraged AI engineering tools (Claude, Windsurf) to accelerate delivery, resolving production code fixes, security flaws, and complex DevOps/pipeline issues.
- Validated core SOC & cybersecurity features, including Threat Dashboards, Security Event Timelines, Investigation Maps, and live monitoring tools.

PULSE SECURE | Sr. Staff Software Test Engineer

Jun 2020 – Oct 2022

San Jose, CA

- Engineered CI/CD pipelines for test automation, improving execution efficiency and reducing test-cycle turnaround.
- Architected a unified master regression suite by integrating UI automation and service-level automation across multiple microservices.
- Customized automation frameworks for dynamic test environments and built a web server to host regression results for tracking and metrics.
- Designed scale tests and automated Zero Trust Access capabilities including certificate validation, certificate rotation, BYOC, and multi-tenancy.
- Created automation to optimize Azure QA/Dev environment utilization and automated Dev/QA cluster deployments using Kubernetes-based internal frameworks.
- Partnered with Sales and Support teams on technical POCs to accelerate customer evaluation and time to market.

FORTINET | Manager, Software Development QA

Dec 2014 – Jun 2020

Sunnyvale, CA

- Defined architecture for high-performance, scalable systems using a security-first strategy and guided software performance improvement initiatives.
- Architected reusable functional automation using Python and Selenium WebDriver and developed a data-driven Python REST API automation framework.
- Established Jenkins CI workflows and automated OVF deployment, build generation, OVF-to-qcow2 conversion, multi-VM vApp deployment, and OpenStack provisioning APIs.
- Developed Python automation for Ixia controllers, recommended product design improvements from performance findings, and directed engineering lab/infrastructure upgrades.
- Recruited and mentored engineers while partnering with Sales, Support, Account teams, and customers on demos, escalations, POCs, and product lifecycle execution.

PAYPAL | Senior Software Engineer

Oct 2013 – Dec 2014

San Jose, CA

- Implemented a Continuous Integration system for a Node.js-based web services team.
- Developed JMeter performance testing and Mocha-based unit/functional regression automation.
- Implemented Node.js and MongoDB web services for Merchant Analytics and automated mobile validation using a Frank/Cucumber-based framework.
- Coordinated localization and accessibility testing and supported production/pre-production deployments.

Recognition: Craftsmanship Award; Spot Award

COLLABORATIVE SOLUTIONS | Software Quality Engineer

May 2013 – Sep 2013

San Jose, CA

- Supported PayPal Here merchant payment services, testing iPhone/iPad applications and automating functional scenarios using the Frank framework.
- Created test scenarios, test cases, and test plans and partnered with development teams on quality status and defect resolution.

DEUTSCHE TELEKOM HBS | QA Engineer

Mar 2011 – May 2013

Mountain View, CA

- Developed automation for e-commerce and billing systems and created JMeter automation for API validation and performance testing.
- Tested Android/iOS applications and validated BSS components supporting ChooChee products and recurring billing.

Recognition: Employee of the Month

EARLIER SOFTWARE & ENGINEERING EXPERIENCE

NXP SEMICONDUCTORS | Sr. Design Engineer

Sep 2007 – Nov 2008

Bangalore, India

- Characterized 45nm libraries and special I/O cells; built tools and Perl scripts to migrate TSMC library data into NXP internal flows and validate CAD standards.

ATRENTA | Sr. Software Engineer

Mar 2006 – Aug 2007

Noida, India

- Developed 1Team Genesis GUI capabilities and reusable Qt/C++ libraries for SoC schematic visualization.

TARAY TECHNOLOGIES | EDA Software Engineer

May 2004 – Apr 2006

Hyderabad, India

- Developed FPGA-based EDA software and tools across MIG007, Virtex-4, Spartan-3, Virtex-2, and Virtex-2 Pro technologies.

EDUCATION

Bachelor of Technology, Electronics & Communication Engineering — DVR College of Engineering / Jawaharlal Nehru Technological University | 2000–2004

CERTIFICATIONS & RECOGNITION

Certifications: Cybersecurity Foundations; Microservices Foundations; Learning the JavaScript Language; Blockchain Basics; Blockchain: Beyond the Basics

Awards: Craftsmanship Award (PayPal); Spot Award (PayPal); Employee of the Month (Deutsche Telekom HBS / ChooChee)